

CPS I — Exercise Sheet 2

University of Freiburg · Winter Term 2025/26

Author: Zhanyu Dong, Xin Pu

1.Implications

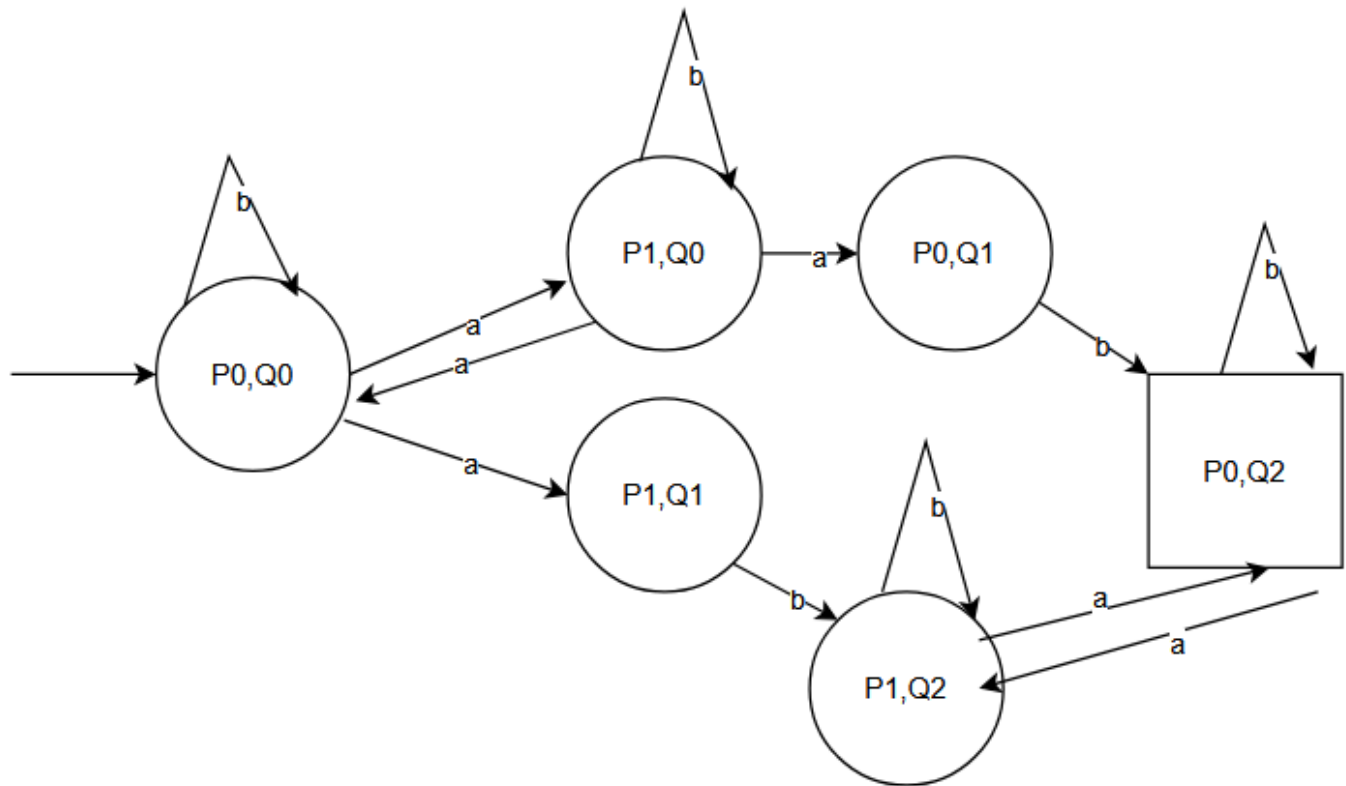
Group 1:

- A implies B
- A is a sufficient condition for B
- A only if B
- A is stronger than B
- $A \wedge \neg B$ does not hold
- $\neg A \vee B$ holds
- If not B, then not A

Group 2:

- B implies A
- A is a necessary condition for B
- A if B
- A is weaker than B

2. Intersection of Finite Automata



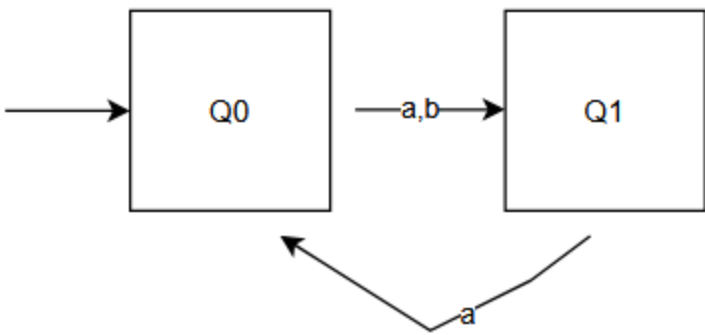
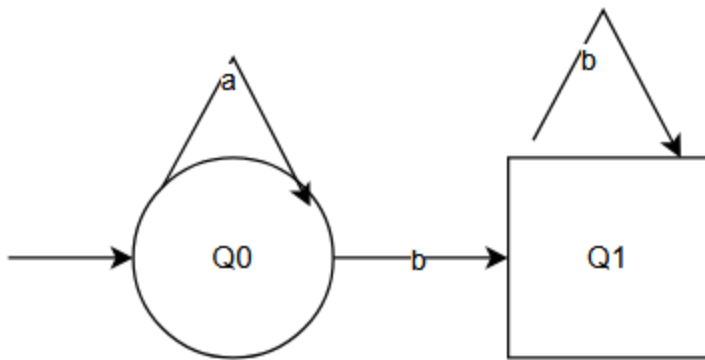
3. Buchi Automata

1. Define

$$L_1 = \{x_0x_1 \dots \mid (\forall i \in \mathbb{N}_0. x_i \in \Sigma) \text{ and } \exists j \in \mathbb{N}_0. \forall i \in \mathbb{N}_0. i > j \text{ and } x_i \neq a\}$$

$$L_2 = \{x_0x_1 \dots \mid (\forall i \in \mathbb{N}_0. x_i \in \Sigma) \text{ and } \forall i \in \mathbb{N}_0. x_{2i+1} = a\}$$

2. Buchi Automata



- **L1**

- $Q: \{Q_0, Q_1\}$
- $\Sigma: \{a, b\}$
- $\delta: \begin{cases} (Q_0, a, Q_0), (Q_0, b, Q_1) \\ (Q_1, b, Q_1) \end{cases}$
- $Q_{\text{init}}: \{Q_0\}$
- $F: \{Q_1\}$

- **L2**

- $Q: \{Q_0, Q_1\}$
- $\Sigma: \{a, b\}$
- $\delta: \begin{cases} (Q_0, a, Q_1), (Q_0, b, Q_1) \\ (Q_1, a, Q_0) \end{cases}$
- $Q_{\text{init}}: \{Q_0\}$
- $F: \{Q_0, Q_1\}$